ARTIFICIAL INTELLIGENCE SOFTWARE DEVELOPMENT

CST8510 Week 9

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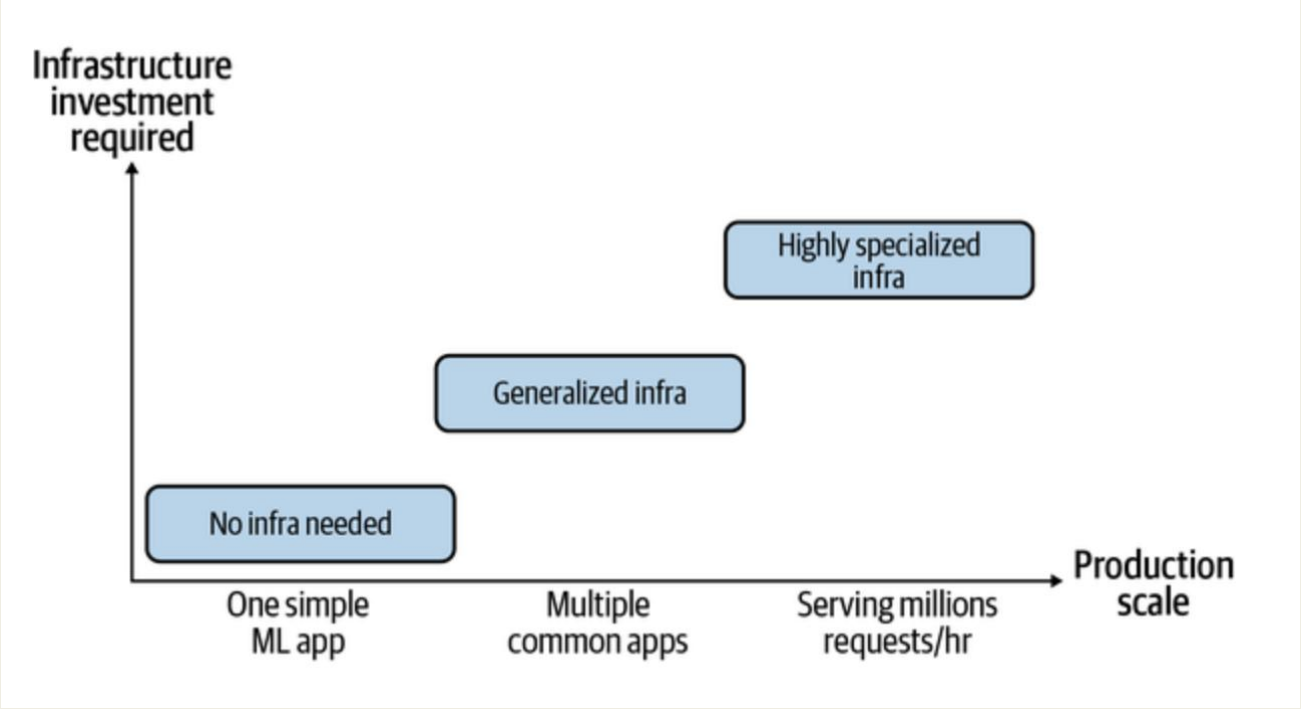
Agenda for Today

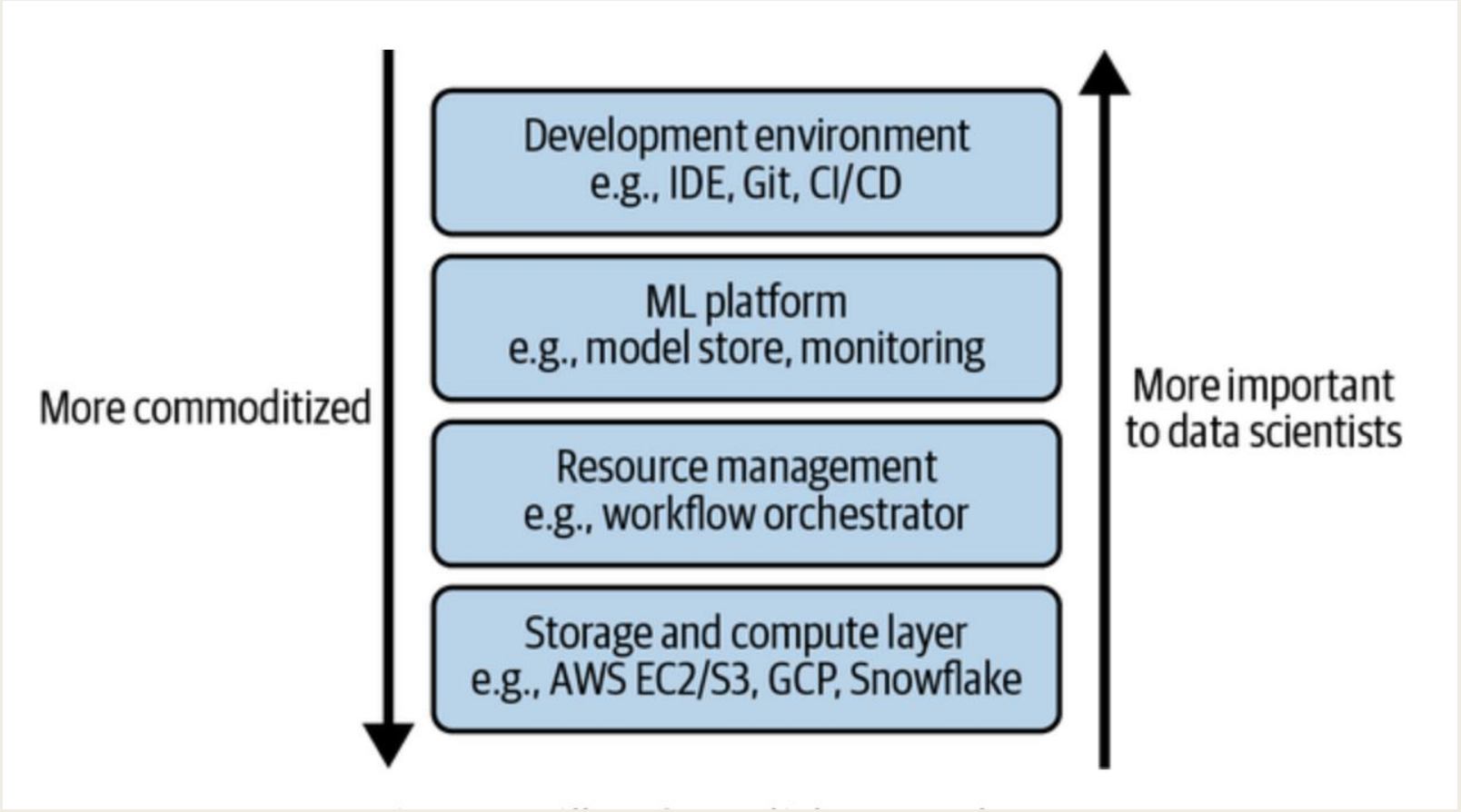
- ❑ Theory: 6:00PM – 8:00PM
  - Right infrastructure for ML Systems
  - Four layers of infrastructure
  - ML Resource Management
  - ML Platform
- ❑ Lab: 8:00PM – 10:00PM
  - Standup Meetings

Infrastructure and Tooling for MLOps

- Infrastructure requirement of different companies.

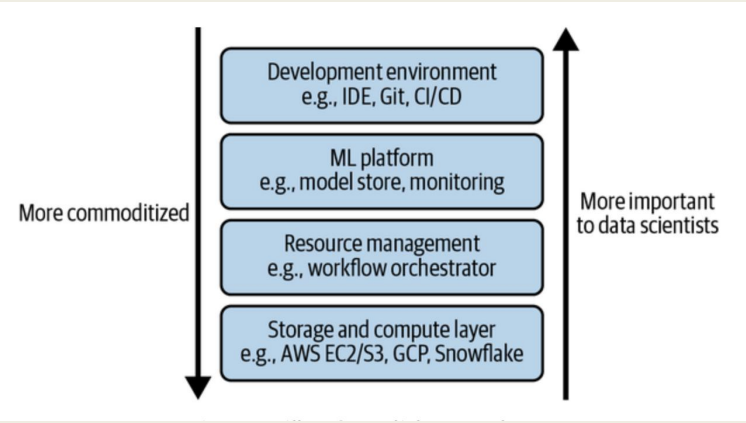


Four Layers of Infrastructure



Four Layers of Infrastructure

- Storage and Compute
  - Layer where data is collected and stored
  - ML workloads are run
- Resource Management
  - Schedule and orchestrate ML workloads
  - Airflow, Kubeflow etc.
- ML Platform
  - Model stores, feature stores and monitoring tools
  - SageMaker, MLFlow
- Development Environment
  - Layer where code is written and experiments are run



Storage and Compute Layer

- Storage layer is where data is stored and collected
- At the basic level storage can be on
  - Hard Drive Disk (HDD)
  - Solid State Drive (SSD)
- Storage layer is completely commoditized and moved to cloud

Storage and Compute Layer

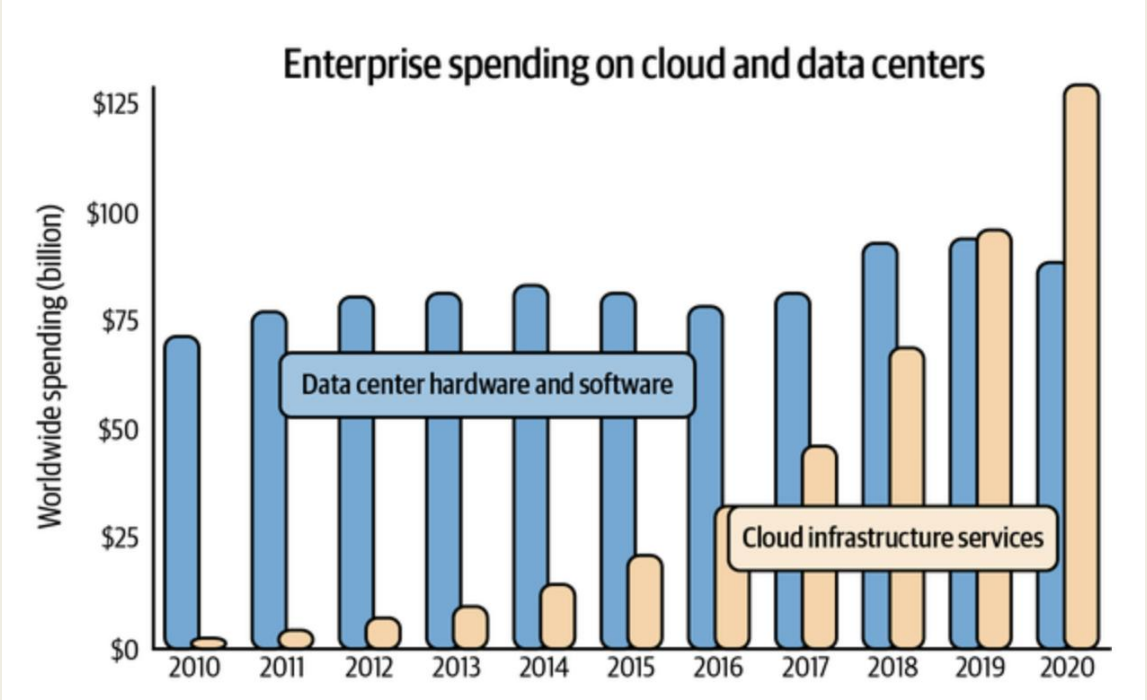
- Compute layer refers to all the compute resources available
- Amount of compute resources available determines the scalability of ML workloads
- The compute layer can usually be sliced into smaller compute units to be used concurrently
  - Threads
  - Containers
  - Pods

Storage and Compute Layer

- Compute units are characterized by two metrics.
  - How much memory it has (GB units)
  - How fast it runs an operation (FLOPS)
- Amount of compute resources available determines the scalability of ML workloads.
- Compute Utilization – Ratio of the number of FLOPS a job can run to the number of FLOPs a compute unit is capable of handling.
- Practically one can only achieve utilization ~ 50%

Storage and Compute Layer

- Cloud spending accounts for approximately 50% cost of revenue of public software companies (analysis by a16z capital venture)
- Some companies are doing "cloud repatriation"



Development Environment

- Environment is where:
  - Code is written
  - Experiments are conducted
  - Interaction with production environment happens

- Dev environment consists of the following components:
  - IDE (Jupyter Notebook, VS Code, Pycharm etc.)
  - Versioning software (Git, DVC, Weights and Biases etc.)
  - CI/CD (Jenkins etc.)

Notebooks

- Notebooks are more than just IDEs, one can include:
  - Images
  - Documentation in LaTeX
  - Other artifacts like tables
- Notebooks are Stateful
  - Retains state after run is completed
  - If program fails, one can restart from where it failed
  - Ideal for doing experiments with large datasets
- Order of execution of the cells is important to keep track

Notebooks

- Companies like Netflix used Notebooks in the production env
- Other tools are developed to run on top of Notebooks
  - **Papermill** - for spawning multiple notebooks with different parameter sets
  - **Commuter** - A notebook hub for viewing, finding, and sharing notebooks within an organization.
  - **nbdev** - a library to write documentation and tests in the same place

Containers

- Production workloads spread out on multiple instances.
- Number of instances dynamically changes upon demand for predictions.
- When a new instance is created, one needs to install dependencies using a list of predefined instructions.
- Container technology is used for this purpose

Docker Containers

- A lightweight, stand-alone, and executable software package.
- Includes everything needed to run a piece of software (code, runtime, system tools, libraries, and settings)
- Containers isolate software from its environment, ensuring that it works consistently across different systems.
- Docker containers are built from images.
- These are templates containing the application's code, runtime, and other dependencies.

Benefits of Docker Containers

- **Portability:** Containers can run on any system with Docker installed, regardless of the underlying infrastructure.
- **Consistency:** Containers ensure that applications behave the same way in development, testing, and production environments.
- **Isolation:** Containers run in their own environment, minimizing conflicts with other applications or system components.
- **Scalability:** Containers can be easily scaled up or down, making it simpler to manage application loads and resources.
- **Version control and sharing:** Docker images can be versioned and shared through repositories like Docker Hub, enabling collaboration and easy updates.

Example of a Docker Container

- ❑ Download the latest PyTorch base image.
- ❑ Clone NVIDIA's apex repository on GitHub, navigate to the newly created *apex* folder, and install apex.
- ❑ Set *fancy-nlp-project* to be the working directory.
- ❑ Clone Hugging Face's transformers repository on GitHub, navigate to the newly created *transformers* folder, and install transformers.

Example of a Docker Container

```
FROM pytorch/pytorch:latest
RUN git clone https://github.com/NVIDIA/apex
RUN cd apex && \
    python3 setup.py install && \
    pip install -v --no-cache-dir --global-option="--cpp_ext" \
    --global-option="--cuda_ext" ./

WORKDIR /fancy-nlp-project
RUN git clone https://github.com/huggingface/transformers.git && \
    cd transformers && \
    python3 -m pip install --no-cache-dir.
```

PODs

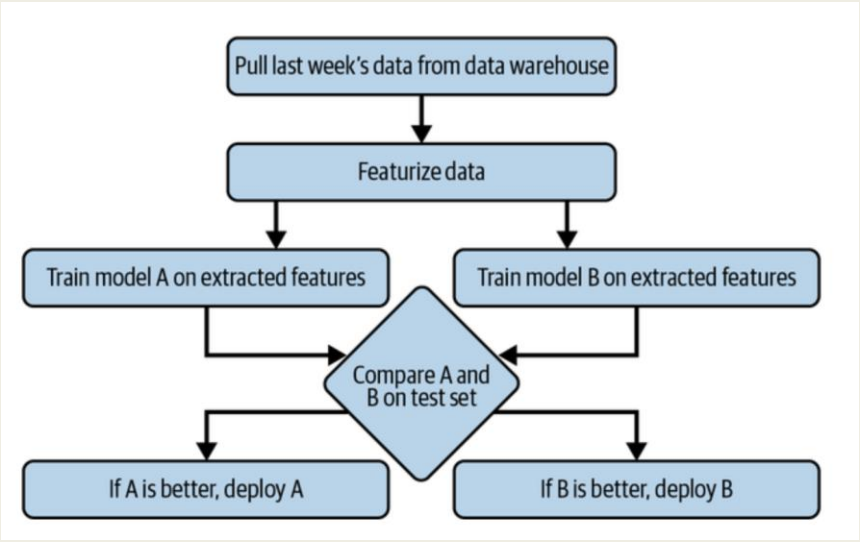
- **Basic Concept:** A pod is a group of one or more containers, with shared storage/network, and a specification for how to run the containers. It is the smallest deployable unit of computing that can be created and managed in Kubernetes.
- **Shared Context:** Containers in the same pod share the same IP address and port space, and can find each other via localhost. They can also share mounted storage.
- **Atomic Unit:** In Kubernetes, the pod is the atomic unit of scaling. When you scale an application up or down, you're actually increasing or decreasing the number of pods.
- **Use Case:** Pods are used when there is a need for a few containers to work together very closely as a single cohesive unit of service.

Resource Management

- In the pre-cloud world resources were limited.
- Focus was then on maximizing resource utilization.
- In the cloud world focus is on using resources cost-effectively.
- Two characteristics of ML workflows that influence their resource management.
  - Repetitiveness
  - Dependencies

Schedulers and Orchestrators

- Cron -.scheduling repetitive jobs to run at fixed times.
- Can not handle the dependencies between the jobs it runs.
- Schedulers are Cron programs that can handle dependencies.
- Takes in the DAG of a workflow and schedules each step accordingly.
- Tend to leverage queues to keep track of jobs
- Need to be aware of the resources available and the resources needed to run each job



Schedulers and Orchestrators

Example of scheduling a job using Slurm

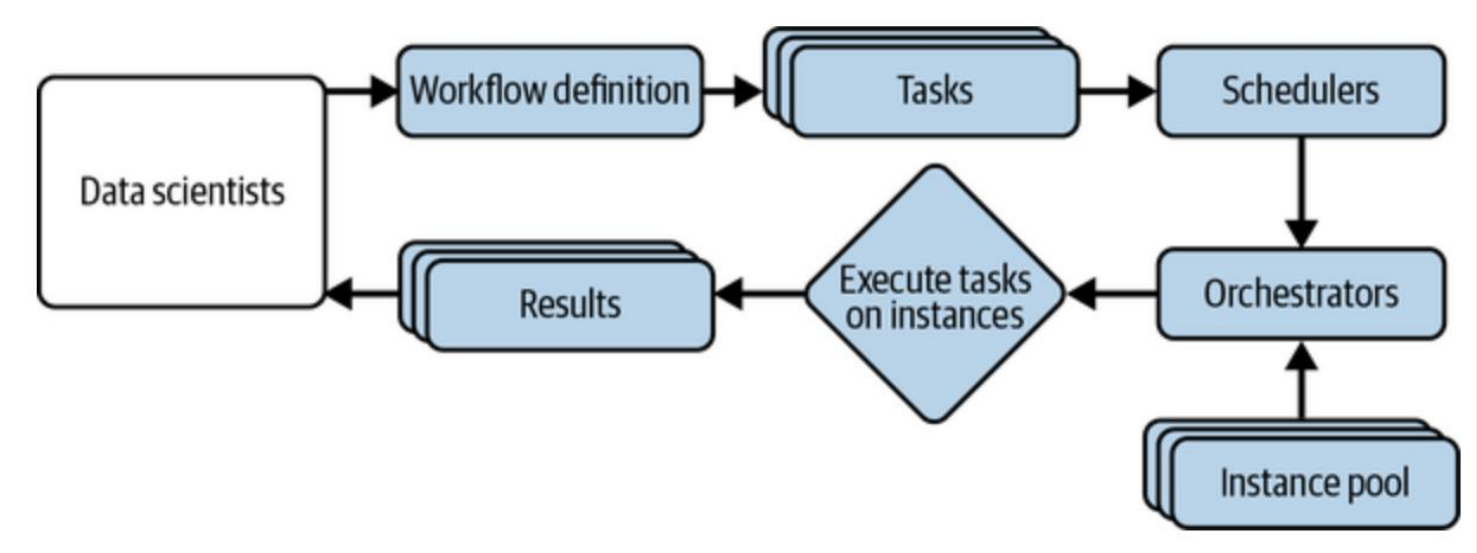
```
#!/bin/bash
#SBATCH -J JobName
#SBATCH --time=11:00:00          # When to start the job
#SBATCH --mem-per-cpu=4096   # Memory, in MB, to be allocated per CPU
#SBATCH --cpus-per-task=4         # Number of cores per task
```

Schedulers and Orchestrators

- Schedulers are concerned with when to run jobs and what resources are needed to run those jobs.
- Orchestrators are concerned with where to get those resources.
- Schedulers deal with job-type abstractions such as DAGs, priority queues, user-level quotas.
- Orchestrators deal with lower-level abstractions like machines, instances, clusters, service-level grouping, replication, etc.
- It can dynamically increase/decrease the number of instances in the available instance pool.
- Most well-known orchestrator today is Kubernetes.

Workflow Management

- They allow you to specify your workflows as DAGs.
- Workflows can be defined using either code (Python) or configuration files (YAML).
- Once a workflow is defined, the underlying scheduler usually works with an orchestrator to allocate resources to run the workflow



Airflow

- One of the first workflow management tool.
- Developed at Airbnb and open sourced.
- Contains a huge library of operators.
- Easy to use with different cloud providers, databases, storage options.

Airflow Drawbacks

- Airflow is monolithic - it packages the entire workflow into one container.
- Airflow's DAGs are not parameterized - you can't pass parameters into your workflows.
- If you want to run the same model with different learning rates, you'll have to create different workflows!
- Airflow's DAGs are static - it can't automatically create new steps at runtime as needed.
- Next generation of workflow orchestrators **Argo** and **Prefect** address these issues.

ML Platform

- ML Platform is a relatively new concept like MLOps
- No universal standard definition exists.
- The shared set of tools for ML deployment makes up the ML platform.
- Most important components:
  - Model Deployment
  - Model Store
  - Feature

ML Platform

- Two important criteria for evaluating the component tools:
  - Whether the tool works with your cloud provider or allows you to use it on your own data center.
  - Need to run and serve models from a compute layer, and usually tools only support integration with a handful of cloud providers.
- Whether it is an open source or a managed service:
  - Opensource tools can be hosted by engineers
  - Less about data security and privacy.
  - More Eng resources are required.
  - Managed services could be more expensive.
  - May not comply with regulations of data storage and privacy.

ML Platform – Model Deployment

- A deployment service helps in both pushing models and their dependencies to production and exposing them as endpoints.
- Deployment is the most mature among all ML platform components.
- All major cloud providers offer tools for deployment:
  - AWS – SageMaker
  - Azure – AzureML
  - GCP – VerTexAI
  - Startups:
    - MLFlow Models:
    - Seldon
    - Cortex
    - Ray-Serve

ML Platform – Model Store

- To help with debugging and maintenance, it is not sufficient to store model object alone
- Information about models stored:
  - Model Definition: Loss function, number of layers of NN, number of parameters in each layer
  - Model Parameters: Actual values of the model parameters after training.
  - Featurize and Predict functions
  - Dependencies: Python packages
  - Data: Pointers to data storage
  - Model Generation Code: Pointers to Github repo
  - Experiment artifacts: Loss curves, performance metrics
  - Tags

ML Platform – Feature Store

- Why Feature store is needed ?
  - Feature management
  - Feature computation
  - Feature consistency
- Popular tools for feature store:
  - Feast:
    - Strong in creating batch features
    - Weak in creating streaming features
  - Tecton:
    - Capable of storing both online and batch features
    - Require deep integration

Summary of Today's Learning

- ❑ Different layers of ML infrastructure.
- ❑ Scheduling and orchestration of different ML tasks.
- ❑ ML resource management
- ❑ Important components of ML platform

Thank You